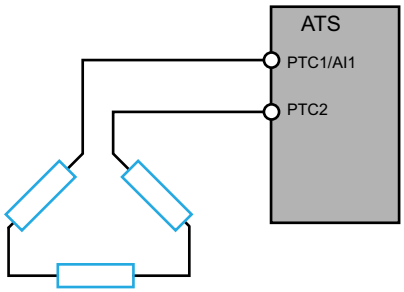
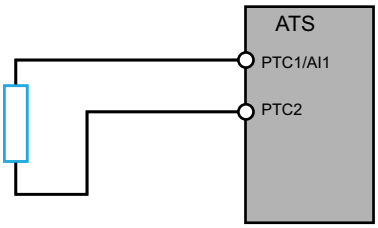
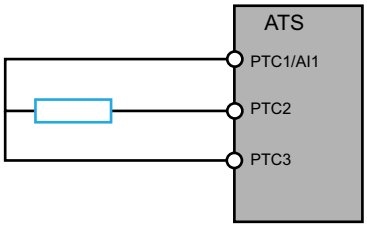
For 2-Wire Sensors

3 serial PTC	Single PTC or PT100
	

For 3-Wire Sensors

Single PT100


Checking Installation

Check List: Before Switching On

Unsuitable settings or unsuitable data or unsuitable wiring may trigger unintended movements, trigger signals, damage parts and disable monitoring functions.

⚠ WARNING

UNANTICIPATED EQUIPMENT OPERATION

- Only start the system if there are no persons or obstructions in the zone of operation.
- Verify that a functioning emergency stop push-button is within reach of all persons involved in the operation.
- Do not operate the product with unknown settings or data.
- Verify that the wiring is appropriate for the settings.
- Never modify a parameter unless you fully understand the parameter and all effects of the modification.
- When commissioning, carefully run tests for all operating states, operating conditions and potential error situations.
- Anticipate movements in unintended directions or oscillation of the motor.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Check List: Mechanical Installation

Verify the mechanical installation of the entire soft starter system:

Step	Action	✓
1	Does the installation meet the specified distance requirements?	
2	Did you tighten all fastening screws according to the specified tightening torque?	

Check List: Electrical Installation

Verify the electrical connections and the wiring:

Step	Action	✓
1	Did you connect all protective Ground conductors?	
2	The correct tightening of the screws may be altered during assembly and wiring phases of the soft starter. Verify and adjust the tightening of all terminal screws to the specified nominal torque.	
3	Do all fuses and circuit breaker have the correct rating; are the fuses of the specified type? Refer to the information provided in the catalog.	
4	Did you connect or insulate all wires at the cable ends?	
5	Did you properly separate and insulate the control and power wiring?	
6	Did you properly connect and install all cables and connectors?	
7	Did you properly connect the signal wires?	
8	Are the required shield connections EMC-compliant?	
9	Did you take all measures for EMC compliance?	
10	Did you confirm that CL1/CL2 terminals are only supplied with 110...230 Vac?	
11	Did you confirm that the output of the relays R1 R2 and R3 are only connected to a maximum voltage of 250Vac / 30Vdc?	

Check List: Covers and Seals

Verify that all devices, doors and covers of cabinet are properly installed to meet the required degree of protection.

Initial Setup

The **[LANGUAGE]** LNG menu is displayed at the first power-up of the ATS480. Refer to the following steps to prepare the ATS480 for the migration procedure.

Step	Action
1	In the **[LANGUAGE]** LNG menu, scroll to the desired device language and press **OK** to validate or press **ESC** to skip this step and keep the labels in English. **Result:** The device labels are now displayed in the selected language.
2	1. In the **[Time Zone]** TOP menu, set the local UTC offset and press **OK** to validate or press **ESC** to skip. 2. In the **[Set Date/Time]** DTO, set the local date and time and press **OK** to confirm or press **ESC** to skip. **Result:** The device is now set on the local time and date.
3	1. In the **[Initial Setup]** ROOT menu, select **[Go to product]** PRDM and press **OK**. 2. Choose a cybersecurity policy: • To set **no credentials** to access this device, refer to step 4. • To set **credentials** or **load an existing cybersecurity policy**, refer to the ATS480 User Manual, page 13.
4	1. Select **[Minimum Cybersec]** CSE and press **OK**. 2. Read the message explaining the functionalities of this profile and press **OK** to validate and access the main menu or **ESC** to cancel the selection. **Result:** The cybersecurity policy is set with no credentials and the device is ready to be commissioned. Refer to Configuration Migration, page 45 to migrate an ATS48 configuration to an ATS480. Selecting the profile **[Minimum Cybersec]** CSE, no credentials will be required to access your process or machine. This setting is saved with the configuration and will be active if a configuration is loaded or copied. **⚠ WARNING** **UNAUTHENTICATED ACCESS AND MACHINE OPERATION** Do not select the profile **[Minimum Cybersec]** CSE if your machine or process is accessible to unauthorized personnel either directly or via a network. **Failure to follow these instructions can result in death, serious injury, or equipment damage.**

Configuration Migration

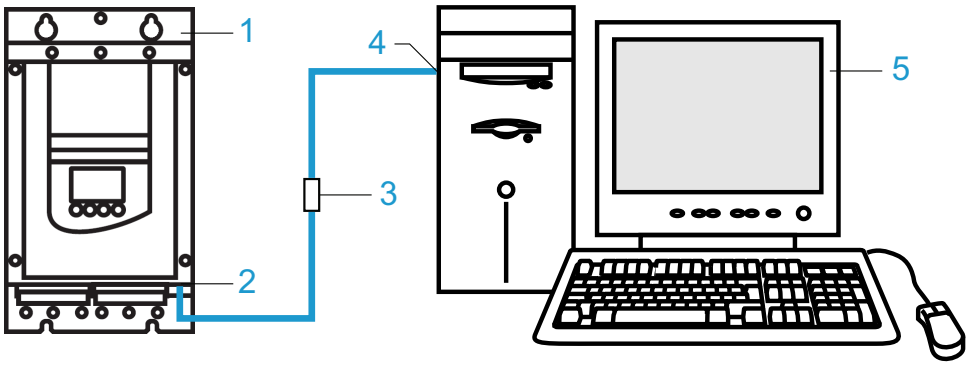
What's in This Part

Requirements 46
Migration Procedure 47
ATS48 Code Equivalence With ATS480 Parameters 51
ATS48 Parameter Changes 54

Requirements

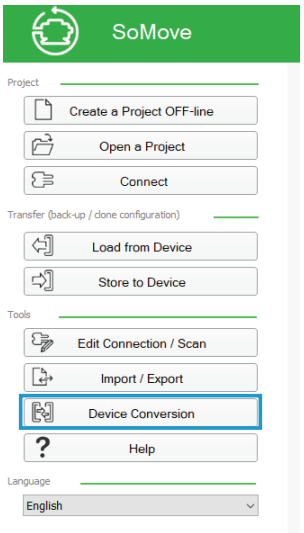
Description	Catalog number and link	
In the menu **[Complete settings]** CST –> **[Command channel]** CCP: • Set the parameter **[Control Mode]** CHCF to **[SE8 Profile]** SE8 (factory setting) This setting is necessary to migrate the ATS48 configuration to the ATS480.		
SoMove setup software Includes: • SoMove setup software for PC in English, French, German, Italian, Spanish and Chinese.	SoMove software can be downloaded from the Schneider Electric website: • *SoMove FDT* (English, French, German, Spanish, Italian, Chinese)	
ATS480 DTM	DTMs (Device Type Managers) can be downloaded from the Schneider Electric website: • DTM: ATS480 DTM Library EN (English — to be installed first), ATS480 DTM Lang FR (French), ATS480 DTM Lang SP (Spanish), ATS480 DTM Lang IT (Italian), ATS480 DTM Lang DE (German), ATS480 DTM Lang CN (Chinese)	
USB/RJ45 cable • Used to connect a PC to the device. • This cable is 2.5 m / 8.20 ft long, has a USB connection (PC end) and an RJ45 connector (device end).	TCSMCNAM3M002P	

Connect the ATS48 to a computer with SoMove installed and power-up the ATS48.

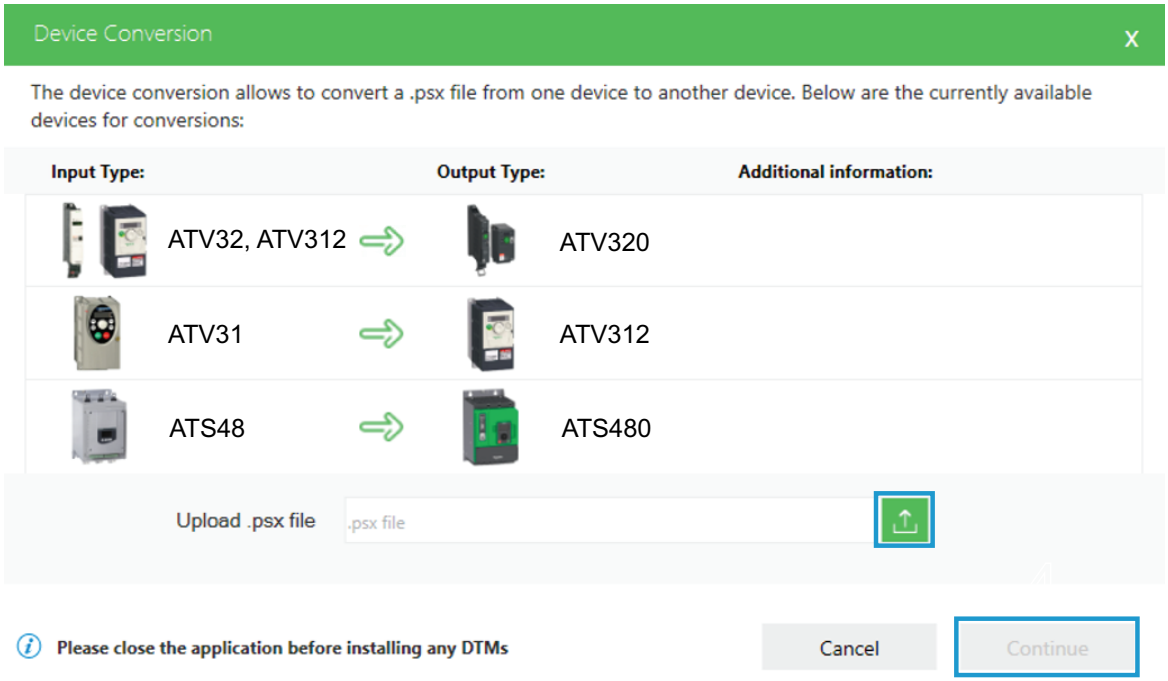
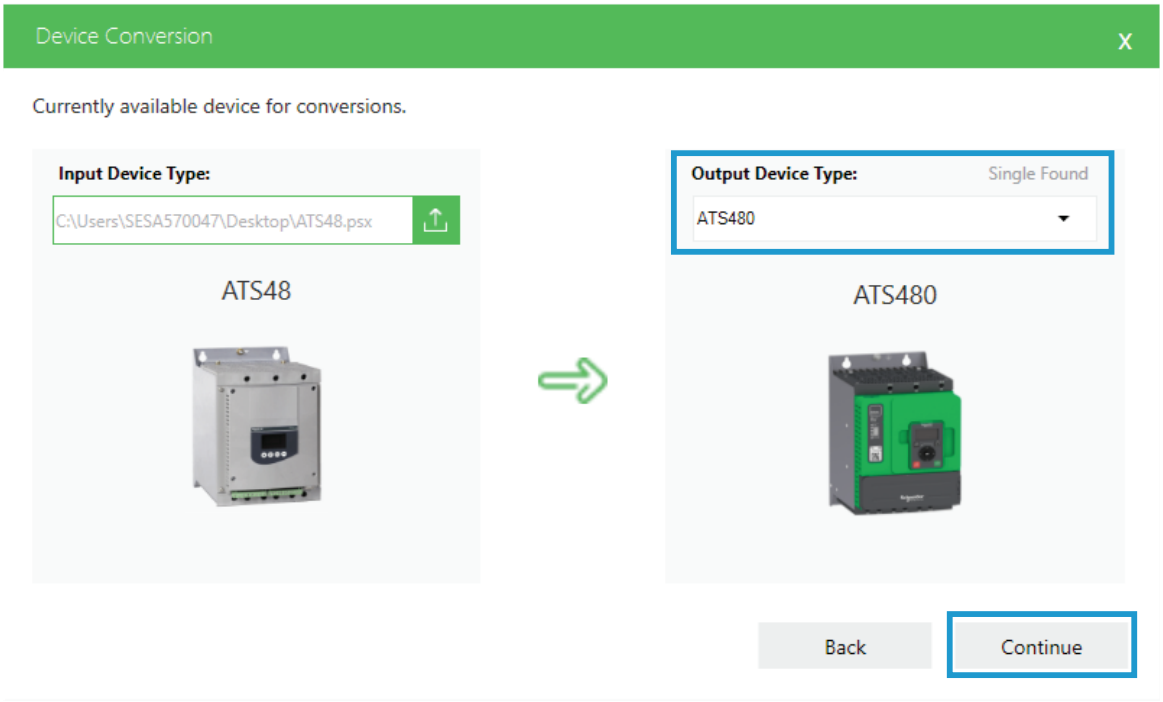


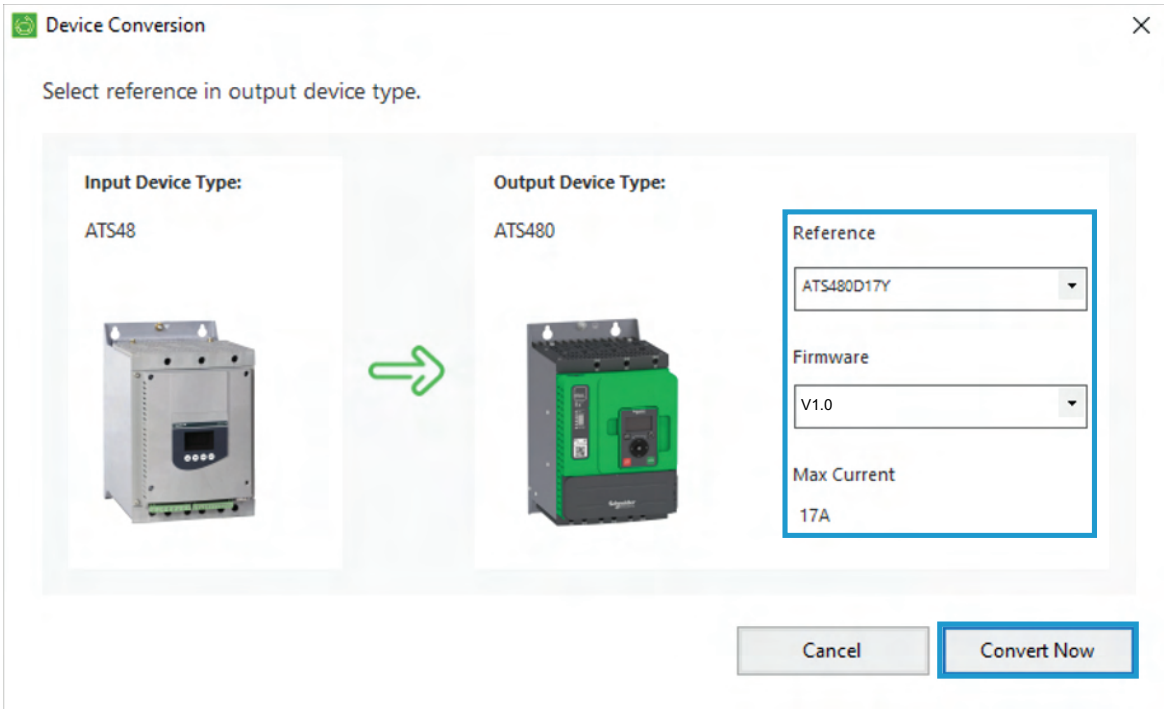
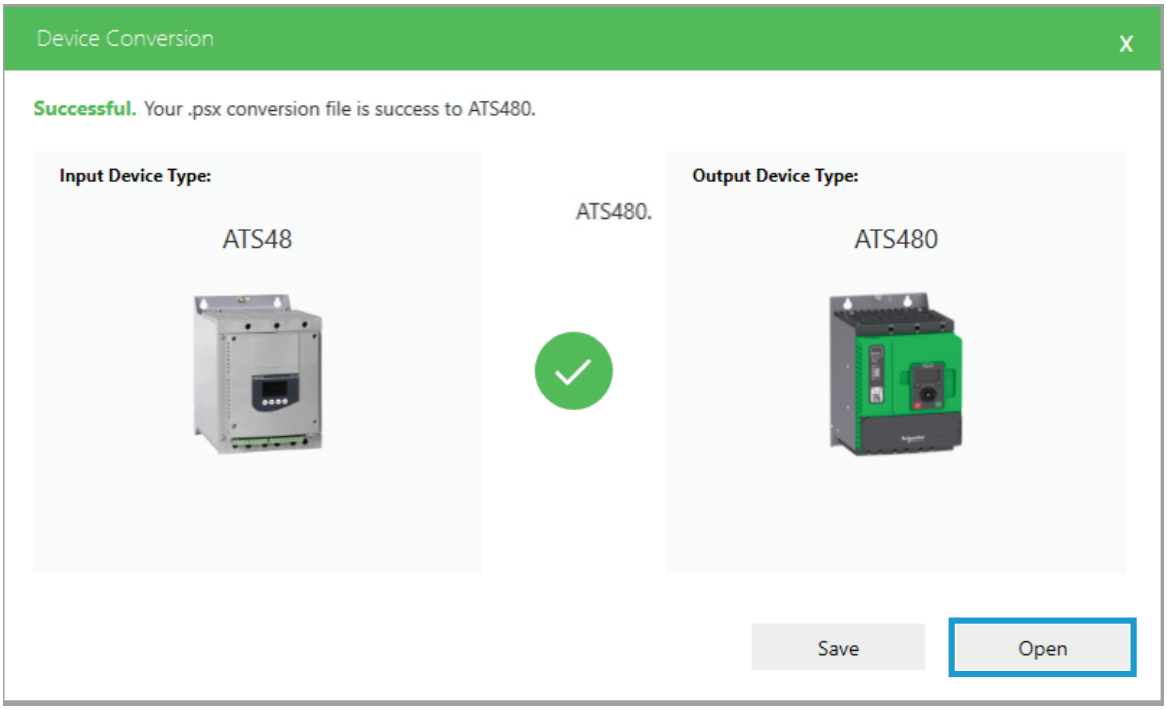
1. ATS48
2. ATS48 Serial Modbus RJ45 port
3. RJ45/USB communication cable TCSMCNAM3M002P
4. PC USB port
5. PC with SoMove installed and the ATS480 DTM

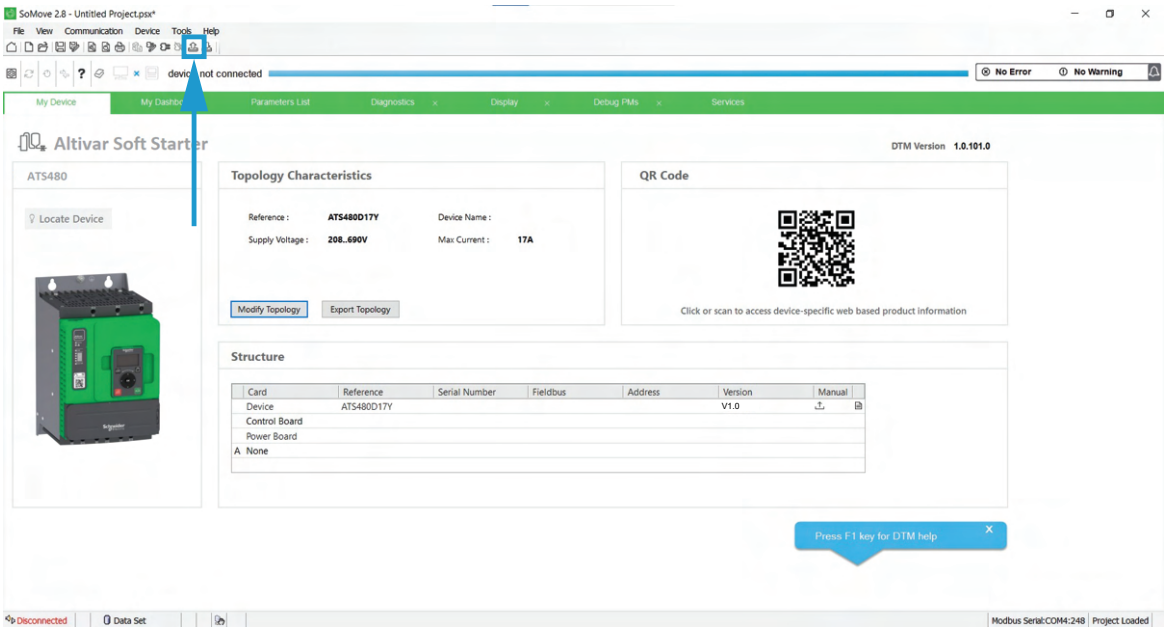
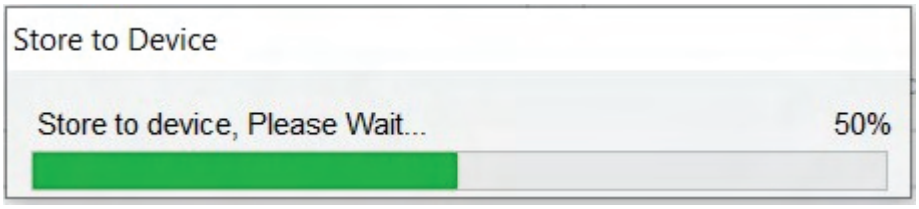
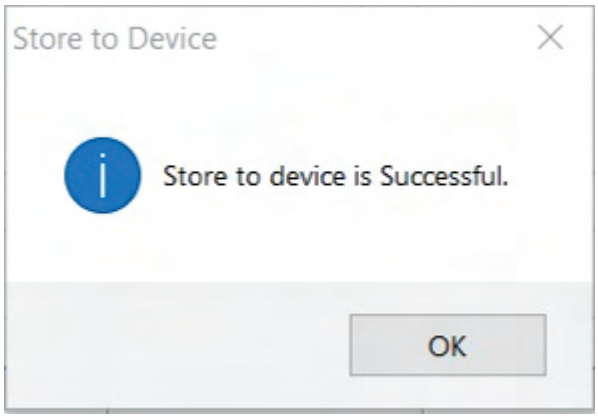
Migration Procedure

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>
1. Connect a computer with SoMove and ATS480 DTM to the ATS48 to substitute. Refer to Requirements, page 46.
2. Start the SoMove application.
3. In the SoMove home page, do one of the following:
• Click the **Device Conversion** button
• Press **CTRL** + **ALT** + **Q** keys, on the menu bar, click **File –> Device Conversion**


Result: Opening of the **Device Conversion** dialog box.
NOTE: The following table shows the default Modbus format:
<table>
<tr><th></th><th>Soft Starter ATS 48</th><th>Soft Starter ATS 480</th></tr>
<tr><td>Default Modbus Format</td><td>8n1</td><td>8e1</td></tr>
</table>
</td></tr>
<tr><td>2</td><td>
In the **Device Conversion** dialog box:
1. Click on
2. Select the configuration file (*.psx*) to convert and click **Open**
3. Click **Continue**
</td></tr>
</table>

Step	Action
	 **Result:** Opening of the **Currently available device for conversions** dialog box
3	In the **Currently available device for conversions** dialog box: 1. Select the target device reference in the **Output Device Type** list 2. Click **Continue**  **Result:** Opening of the **Select reference in output device type** dialog box

Step	Action
4	In the **Select reference in output device type** dialog box: 1. Select the output device reference 2. Select the latest output device firmware version available 3. Check the output device current rating 4. Click **Convert Now**  **Result:** Opening of the list of converted parameters and their value in the ATS480 project
5	In the **Successfully psx file conversion** dialog box: 1. Check the input and output devices 2. Click **Open**  **Result:** Opening of the new ATS480 project with the converted .psx file

Step	Action
6	In the toolbar click **Store to Device**  **Result:** Opening of the **Store to Device** dialog box, loads the converted .psx file to the output device 
7	If the transfer is successful: **Result:** Opening of the **Store to device is Successful** dialog box  If the transfer is not successful, verify the connection with the output device. When the transfer is successful, restart the device to take the new Modbus communication values into account. This conclude the configuration transfer from the ATS48 to the ATS480. To modify the migrated configuration, refer to the ATS480 User Manual, page 13.

ATS48 Code Equivalence With ATS480 Parameters

This table presents the equivalence between the ATS48 codes and the ATS480 parameters visible on the display terminal.

ATS48		ATS480	
Code	**HMI access path**	**HMI access path**	**Parameter**
SEt	Main menu	1 **[Simply Start]** SYS ➡ **[Simply start]** SIM	**[Simply Start]** SYS
In	SEt	1 **[Simply Start]** SYS ➡ **[Simply start]** SIM	**[Motor Nom Current]** IN
ILt	SEt		**[Current Limit]** ILT
uLn	drC		**[Mains Voltage]** ULN
ACC	SEt		**[Acceleration]** ACC
tq0	SEt		**[Init Starting Torque]** TQ0
Sty	SEt		**[Type Of Stop]** STY
dEC	SEt		**[Deceleration]** DEC
EdC	SEt		**[End Of Deceleration]** EDC
brC	SEt		**[Braking Level]** BRC
EbA	SEt		**[DC Braking Time]** EBA
Pro	Main menu	Main menu	**[Monitoring]** PROT
tHP	Pro	2 **[Monitoring]** PROT	**[Motor Class]** THP
uLL	Pro	2 **[Monitoring]** PROT ➡ **[Process underload]** ULD	**[Underload Activation]** UDLA
tuL	Pro		**[Unld Detect Delay]** ULT
LuL	Pro		**[Unld.Thr.0.Speed]** LUL
uLL	Pro		**[Underload ErrorResp]** UDL
tLS	Pro	2 **[Monitoring]** PROT	**[Too Long Start]** TLS
oIL	Pro	2 **[Monitoring]** PROT ➡ **[Process overload]** OLD	**[Overload Activation]** ODLA
toL	Pro		**[Ovld Detection Delay]** TOL
LoC	Pro		**[Overload Threshold]** LOC
oIL	Pro		**[Overload ErrorResp]** ODL
PHr	Pro	2 **[Monitoring]** PROT	**[Phase Inversion Mon]** PHR
tbS	Pro		**[Time Before Restart]** TBS
PHL	Pro		**[Phase Loss Cur Thd]** PHL
rtH	Pro		**[Mot Th State Reset]** RTHR
PtC	Pro	2 **[Monitoring]** PROT ➡ **[Thermal monitoring]** TPP	**[AI1 Th Monitoring]** TH1S **[AI1 Type]** AI1T **[AI1 Th Error Resp]** TH1B
drC	Main menu	Main menu	**[Complete settings]** CST
FrC	drC	3 **[Complete settings]** CST ➡ **[Motor parameters]** MPA	**[Mains Frequency]** FRC
dLt	drC		**[Inside Delta]** DLT
SSt	drC		**[Small Motor Test]** SST

ATS48		ATS480	
Code	**HMI access path**	**HMI access path**	**Parameter**
bSt	drC	3 **[Complete settings]** CST ➡ **[Motor wiring]** MWMT	**[Boost]** BST **[Init Starting Voltage]** V0
CLP	drC		**[Control Mode]** CLP
iPr	io	3**[Complete settings]** CST ➡ **[Preheating]** PRF	**[Preheat Level]** IPR
tPr	io		**[Time Before Preheat]** TPR
tiG	drC	3 **[Complete settings]** CST ➡ **[Start & Stop]** SSP	**[Deceleration Gain]** TIG
tLi	drC		**[Torque Limit]** TLI
LSC	drC		**[Stator Loss Comp]** LSC
CSC	drC	3 **[Complete settings]** CST ➡ **[Cascade]** CSC	**[Cascade Activation]** CSC
ArS	Pro	3 **[Complete settings]** CST ➡ **[Error/Warning handling]** CSWM	**[Auto Fault Reset]** ATR
io	Main Menu	Main Menu	**[Input/Output]** IO
Li3	io	4 **[Input/Output]** IO	**[DI3 High Assignment]** L3H
Li4	io		**[DI4 High Assignment]** L4H
Lo1	io	4 **[Input/Output]** IO ➡ **[DQ1 configuration]** DO1	**[DQ1 Assignment]** DO1
Lo2	io	4 **[Input/Output]** IO ➡ **[DQ2 Configuration]** DO2	**[DQ2 Assign]** DO2
Ao	io	4 **[Input/Output]** IO ➡ **[AQ1 configuration]** AO1	**[AQ1 assignment]** AO1
O4	io		**[AQ1 Type]** AO1T **[AQ1 min output]** AOL1 **[AQ1 max output]** AOH1
ASC	io		**[AQ1 Scaling]** AO1S
r1	io	4 **[Input/Output]** IO ➡ **[R1 configuration]** R1	**[R1 Assignment]** R1
r3	io	4 **[Input/Output]** IO ➡ **[R3 configuration]** R3	**[R3 Assignment]** R3
St2	Main menu	Main menu	**[2nd Mot Parameters]** ST2
in2	St2	5 **[2nd Mot Parameters]** ST2	**[Nom Current Motor 2]** INM2
iL2	St2		**[Current Limit Motor 2]** ILM2
AC2	St2		**[Acceleration Motor 2]** ACM2
tq2	St2		**[Init Start Torque Mot 2]** TQM2
dE2	St2		**[Deceleration Motor 2]** DEM2
Ed2	St2		**[End Of Dec Motor 2]** EDM2
tL2	St2		**[Torque Limit Motor 2]** TLM2
ti2	St2		**[Dec Gain Motor 2]** TIM2
CoP	Main menu	Main menu	**[Communication]** COM

ATS48		ATS480	
Code	**HMI access path**	**HMI access path**	**Parameter**
Add	CoP	6 **[Communication]** COM ➡ **[Modbus Fieldbus]** MD1	**[Modbus Address]** ADD
tbr	CoP		**[Modbus Baud Rate]** TBR
For	CoP		**[Modbus Format]** TFO
tLP	CoP		**[ModbusTimeout]** TTO
SuP	Main menu	Main menu	**[Display]** MON
CoS	SuP	7 **[Display]** MON ➡ **[Motor parameters]** MMO	**[Power Factor]** COS
LCr	SuP		**[Motor Current]** LCR
LPr	SuP		**[Acv Elc Out Pwr in %]** EPR
Ltr	SuP		**[Motor Torque]** LTR
PHE	SuP		**[Phase Direction]** PHE
tHr	SuP	7 **[Display]** MON ➡ **[Thermal Monitoring]** TPM	**[Motor Therm State]** THR
rPr	drC	7 **[Display]** MON ➡ **[Counter Management]** ELT	**[Counter Reset]** RPR
rnt	SuP	7 **[Display]** SUP ➡ **[Counter Management]** ELT	**[Motor Run Time]** RTHH
LAP	SuP	7 **[Display]** MON ➡ **[Energy parameters]** ENP	**[Acv Elc Out Pwr in kW]** EPRW
LFt	SuP	8 **[Diagnostics]** DIA ➡ **[Diag. data]** DDT	**[Last Error]** LFT
FCS	drC	9 **[Device Management]** DMT ➡ **[Factory settings]** FCS	**[Factory settings]** FCS
EtA	SuP	Not migrated.	
Cod	SuP	Not migrated	

ATS48 Parameter Changes

Compatible

The following parameters has been modified from ATS48 to ATS480.

- The soft starter behavior is identical between ATS48 and ATS480
- The following changes do not affect the migration procedure
- To access to all parameters, set **[Access Level]** LAC in the **[My preferences]** MYP menu to **[Expert]** EPR.

0 4 – Configuration of the type of signal supplied by output AO

The 0 4 ATS48 parameter set the type of signal supplied by the control terminal AO1:

- 020: 0 – 20 mA signal
- 040: 4 – 20 mA signal

For ATS480 this functionality is split into the 3 following parameters:

- **[AQ1 Type]** AO1T for setting the type of signal supplied by the terminal AQ1
- **[AQ1 min output]** AOL1 for setting the lower limit of the current output from AQ1
- **[AQ1 max output]** AOH1 for setting the upper limit of the current output from AQ1

With ATS48 the scaling of the value measured by AO1 is assured by the parameter A S C. This functionality is identical for ATS480 with the parameter **[AQ1 Scaling]** AO1S.

ATS48 parameters

Code	Setting	Factory setting
0 4	020 – 420	020
• 020: 0 – 20 mA signal • 040: 4 – 20 mA signal		
A S C	50...500%	200
• Motor current scaling: A S C percentage multiplied by the nominal motor current • Motor torque scaling: A S C percentage multiplied by the nominal motor torque • Motor thermal state: A S C percentage multiplied by 100% • Power factor: A S C between 0 and 1. • Motor electrical active power: A S C percentage multiplied by the nominal motor power		

ATS480 parameters

Parameter	Setting	Factory setting
[AQ1 Type] AO1T	–	**[Current]** 0A
AQ1 Type This parameter set the type of analogue output. • **[Voltage]** 10U: voltage output • **[Current]** 0A: current output Access path: **[Input/Output]** ➡ **[AI/AQ]** ➡ **[AQ1 configuration]**		
[AQ1 min output] AOL1	0.0...20.0 mA	0.0 mA
AQ1 min output value This parameter set the minimum current output from AQ1 This parameter is visible only if **[AQ1 Type]** AO1T is set to **[Current]** 0A. Access path: **[Input/Output]** ➡ **[AI/AQ]** ➡ **[AQ1 configuration]**		
[AQ1 max output] AOH1	0.0...20.0 mA	20.0 mA
AQ1 max output value This parameter set the maximum current output from AQ1 This parameter is visible only if **[AQ1 Type]** AO1T is set to **[Current]** 0A. Access path: **[Input/Output]** ➡ **[AI/AQ]** ➡ **[AQ1 configuration]**		
[AQ1 Scaling] AO1S	50...500%	200
Analog output AQ1 scaling • Motor current scaling: **[AQ1 Scaling]** percentage multiplied by the nominal motor current • Motor torque scaling: **[AQ1 Scaling]** percentage multiplied by the nominal motor torque • Motor thermal state: **[AQ1 Scaling]** percentage multiplied by 100% • Power factor: **[AQ1 Scaling]** between 0 and 1. • Motor electrical active power: **[AQ1 Scaling]** percentage multiplied by the nominal motor power • Access path: **[Input/Output]** ➡ **[AI/AQ]** ➡ **[AQ1 configuration]**		

CLP – Torque control

The CLP ATS48 parameter set the torque control to On or OFF.

The ATS480 parameter **[Control Mode]** CLP behave identically, the difference being the name of the settings:

- On is changed to **[Torque Control]** TC
- OFF is changed to **[Voltage Control]** VC

ATS48 parameter

ATS48 Code	Setting	Factory setting
CLP	On – OFF	On
• On: Torque control active • OFF: Torque control inactive When the torque control is inactive, acceleration and deceleration are controlled by voltage variation.		

ATS480 parameter

Parameter	Setting	Factory setting
[Control Mode] CLP	–	**[Torque Control]** TC
Control mode • **[Torque Control]** TC: enable torque control. • **[Voltage Control]** VC: enable voltage control Access path: **[Complete settings]** ➡ **[Start & Stop]**		

bSt – Voltage boost level

With **[Control Mode]** CLP set to **[Torque Control]** TC (factory setting):

- The ATS480 **[Boost]** BST parameter is identical to the ATS48 bSt parameter.

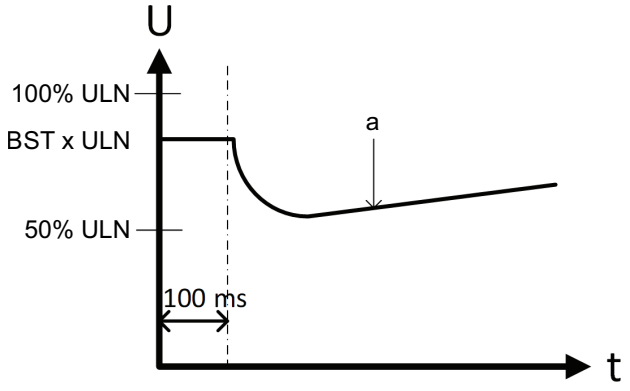
ATS48 and ATS480 parameter

ATS48 Code	Setting	Factory setting
bSt	50...100% or OFF	OFF

The ATS48 bSt parameter provide a boost of 100 ms at the start to overcome a mechanical hard point. The level of the starting boost can be set between 50% and 100% of **[Mains Voltage]** ULN. At the end of boost, the starting ramp follow the start profile set by ACC and tq0.

The ATS480 **[Boost]** BST parameter is identical.

- OFF: Function inactive
- 50%...100%: setting as a % of nominal motor voltage during boost



- U: Voltage
- t: Time
- a: Voltage generated by the torque control
- ULN: **[Mains Voltage]** ULN, mains supply set in **[Simply Start]** SYS.

NOTE: Setting the value of this parameter too high can cause overcurrent and trigger error such as **[Overcurrent]** OCF

With **[Control Mode]** CLP set to **[Voltage Control]** VC:

- The ATS48 bSt parameter can be set between 25% and 100% of **[Mains Voltage]** ULN. This range regroups two different behaviors:
  - Between 25% and 49%, bSt set the initial voltage of the starting ramp
  - Between 50% and 100%, bSt set the level of the starting boost for 100 ms before following the starting ramp

To clarify, the bSt parameter has been split into the 2 following parameters:

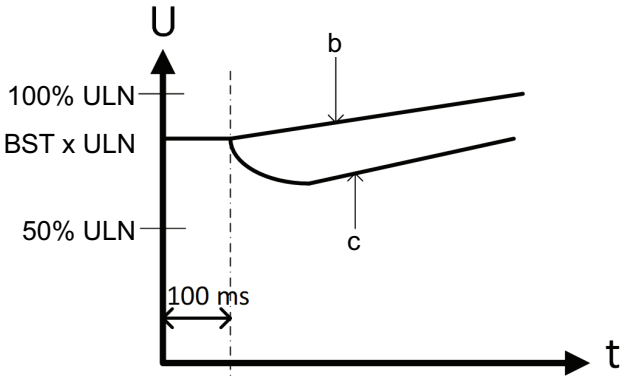
- **[Boost]** BST restricted between 50% and 100% which set the level of the boost for 100 ms before following the starting ramp
- **[Init Starting Voltage]** V0 restricted between 25% and 49% which set the initial voltage of the starting ramp

When **[Boost]** BST is active, **[Init Starting Voltage]** V0 is ignored and not visible.

ATS480 parameter – Boost with voltage control

Parameter	Setting range	Factory setting
[Boost] BST	50...100% of **[Mains Voltage]** ULN or **[No]** NO	**[No]** NO

When **[Control Mode]** CLP is set to **[Voltage Control]** VC the starting ramp initialized with **[Boost]** BST follow this curve:



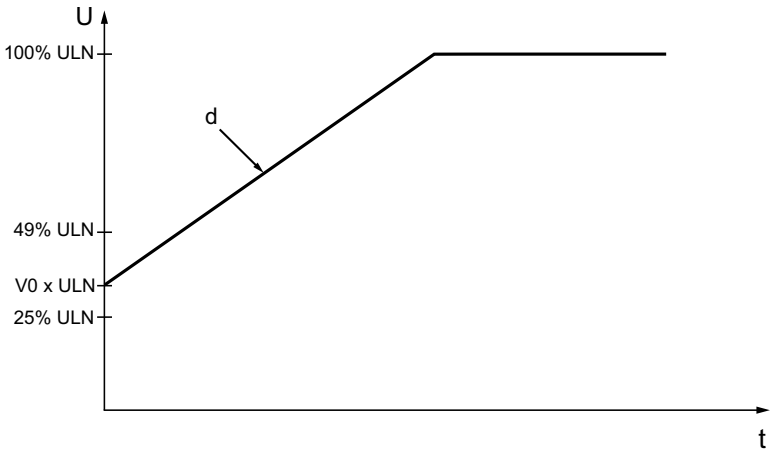
- U: Voltage
- t: Time
- b: Voltage ramp initialized to **[Boost]** BST level
- c: Voltage ramp in case of current limitation
- ULN: **[Mains Voltage]** ULN, mains supply set in **[Simply Start]** SYS.

Access path: **[Complete settings]** ➡ **[Start & Stop]**

ATS480 parameters – Initial voltage

Parameter	Setting range	Factory setting
[Init Starting Voltage] V0	25...49% of **[Mains Voltage]** ULN	49%

Set the initial voltage of the starting ramp.



- U: Voltage
- t: Time
- d: Voltage starting ramp

This parameter is visible if:

- **[Control Mode]** CLP is set to **[Voltage Control]** VC
- **[Boost]** BST is set to **[No]** NO

Access path: **[Complete settings]** ➡ **[Start & Stop]**

o i L – Activation of current overload

The o i L ATS48 parameter enables the motor overload monitoring and the soft starter behavior when the measured motor current exceeds the set threshold:

For ATS480 the activation of this function is splitted into the 2 following parameters:

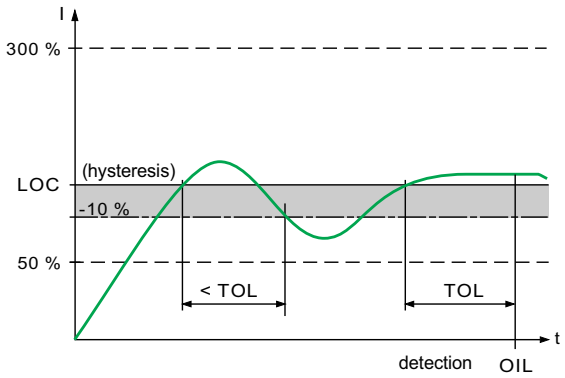
- **[Overload Activation]** ODLA to enable or disable the motor overload monitoring
- **[Overload ErrorResp]** ODL to set the soft starter behavior when the measured motor current exceeds the set threshold. The definition of the threshold and detection time is same as ATS48.

ATS48 parameter

ATS48 Code	Setting	Factory setting
o i L	–	OFF

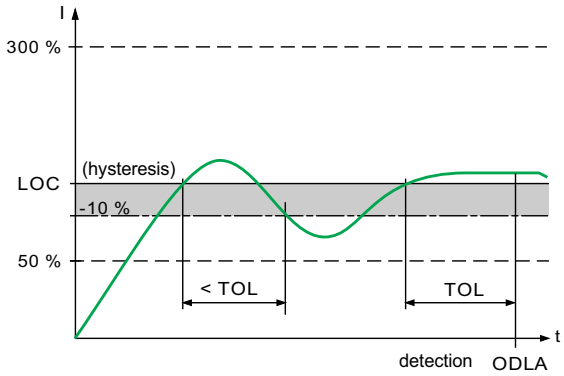
If the motor current exceeds an adjustable threshold L o C for a period of time longer than an adjustable value t o L:

- A L A: an alarm is activated (internal bit and configurable logic output)
- d E F: the soft starter is locked and the o L C fault is triggered
- o F F: deactivate this functionality



- I: Current
- t: Time

ATS480 parameters

Parameter	Setting	Factory setting
[Overload Activation] ODLA	–	**[No]** NO
Overload activation This parameter enable overload monitoring. The monitoring is active when the soft starter displays **[Running]** RUN. If the motor current exceeds the threshold set in **[Overload Threshold]** LOC for a duration longer than the value set in **[Ovld Detection Delay]** TOL, the soft starter behave according to the value set in **[Overload ErrorResp]** ODL. • **[Yes]** YES: Activate overload monitoring • **[No]** NO: Deactivate overload monitoring  • I: Current • t: Time Access path: **[Monitoring]** ➡ **[Process overload]**		
[Overload ErrorResp] ODL	–	**[No]** NO
Response to overload error This parameter set the soft starter behavior when the motor current exceeds the threshold set in **[Overload Threshold]** LOC for a duration longer than the value set in **[Ovld Detection Delay]** TOL. • **[No]** NO: trigger a warning (internal bit and configurable digital output) • **[Yes]** YES: trigger the **[Process Overload]** OLC error This parameter can be accessed if **[Overload Activation]** ODLA is set to **[Yes]** YES. Access path: **[Monitoring]** ➡ **[Process overload]**		

uLL – Activation of motor underload

The uLL ATS48 parameter enables the motor underload monitoring and the soft starter behavior when the measured motor torque is lower than the set threshold:

For ATS480 the activation of this function is splitted into the 2 following parameters:

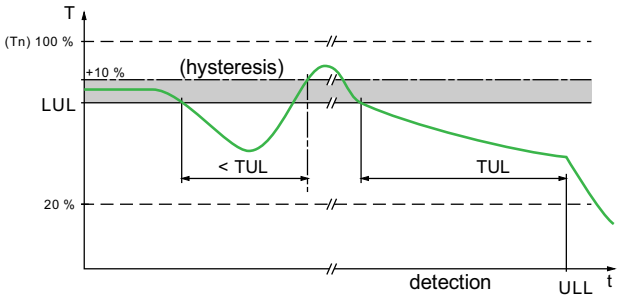
- **[Underload Activation]** UDLA to enable of disable the motor underload monitoring
- **[Underload ErrorResp]** UDL to set the soft starter behavior when the measured torque is below the set threshold. The definition of the threshold and detection time is same as ATS48.

ATS48 parameter

ATS48 Code	Setting	Factory setting
uLL	–	OFF

If the motor torque is less than an adjustable threshold LuL for a period of time longer than an adjustable value tuL:

- ALA: an alarm is activated (internal bit and configurable logic output)
- dEF: the soft starter is locked and the uLF fault is triggered
- oFF: deactivate this functionality



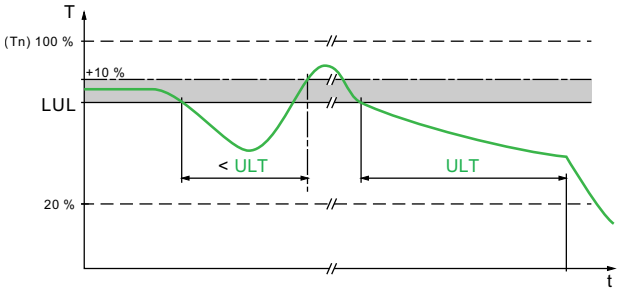
- T: Torque
- t: Time

ATS480 parameters

Parameter	Setting	Factory setting
[Underload Activation] UDLA	**[Yes]** YES or **[No]** NO	**[No]** NO

Underload activation
This parameter enable underload monitoring.

If the motor torque is inferior to the threshold set in **[Underload Threshold]** LUL for a duration longer than the value set in **[Unld Detect Delay]** ULT, the soft starter behave according to the value set in **[Underload ErrorResp]** UDL.



- T: Torque
- t: Time

ATS480 parameters (Continued)

Access path: **[Monitoring]** ➡ **[Process underload]**		
[Underload ErrorResp] UDL	**[Yes]** YES or **[No]** NO	**[OFF]** OFF
Response to underload error This parameter set the soft starter behavior when the motor torque is inferior to the threshold set in **[Underload Threshold]** LUL for a duration longer than the value set in **[Unld Detect Delay]** ULT. • **[Yes]** YES: trigger the **[Process Underload]** ULF error • **[No]** NO: trigger a warning (internal bit and configurable digital output) This parameter can be accessed if **[Underload Activation]** UDLA is set to **[Yes]** YES. Access path: **[Monitoring]** ➡ **[Process underload]**		

PtC – Activation of motor monitoring by PTC probes

The PtC ATS48 parameter provides motor thermal monitoring with PTC probes and set the soft starter behavior when the measured motor temperature trigger the warning otF:

ATS48 parameter

ATS48 Code	Setting	Factory setting
PtC	–	OFF
The PTC probes on the motor must be connected to the AI1 input on the soft starter. This monitoring is independent of the calculated thermal protection (tHP parameter). Both types of protection can be used simultaneously. • ALA: an alarm is activated (internal bit and configurable logic output) • dEF: the soft starter is locked and the otF error is trigered • oFF: deactivate this functionality		

Because the ATS480 allows the use of PTC and PT100 probes this functionality is splitted into the 5 following parameters:

- **[AI1 Th Monitoring]** TH1S to enable or disable the thermal probe monitoring on the terminal AI1
- **[AI1 Type]** AI1T to set the type of thermal probe used on AI1
- **[AI1 Th Error Resp]** TH1B to set the soft starter behavior when the AI1 monitoring triggers an error
- **[AI1 Th Error Level]** TH1F
- **[AI1 Th Warn Level]** TH1A

ATS480 parameters

Parameter	Setting	Factory setting
[AI1 Th Monitoring] TH1S	–	**[No]** NO
Activation of the thermal monitoring on AI1 This parameter activates the thermal monitoring with thermal probes on AI1 • **[Not Configured]** NO: Thermal monitoring on PTC1/AI1 disabled • **[AI1]** AI1: Thermal monitoring on PTC1/AI1 enabled Access path: **[Monitoring]** ➡ **[Thermal monitoring]**		
[AI1 Type] AI1T	–	**[PTC]** PTC
Configuration of AI1 This parameter sets the type of probe on AI1. • **[PTC]** PTC: PTC • **[PT100]** 1PT2: PT100 • **[PT100 in 3 wires]** 1PT23: PT100 in 3 wires This parameter is visible only if **[AI1 Th Monitoring]** TH1S is **not** set to **[Not Configured]** NO. Access path: **[Monitoring]** ➡ **[Thermal monitoring]**		
[AI1 Th Error Resp] TH1B	–	**[Freewheel Stop]** YES
Response to thermal error for AI1 This parameter sets the behavior of the soft starter when an error is triggered by AI1 input. • **[Ignore]** NO: Soft starter ignore the error • **[Freewheel Stop]** YES: Error is triggered and motor stop in freewheel • **[Configured Stop]** STT: Error is triggered and motor stop according to the value set in **[Type of stop]** STT Access path: **[Monitoring]** ➡ **[Thermal monitoring]**		
[AI1 Th Error Level] TH1F	–15°C/5°F...200°C/392°F	110°C/230°F
Thermal error level for AI1 This parameter sets the temperature threshold to trigger the **[AI1 Th Level Error]** TH1F. This parameter is visible when **[AI1 Th Monitoring]** TH1S is set to **[AI1]** AI1. Access path: **[Monitoring]** ➡ **[Thermal monitoring]**		
[AI1 Th Warn Level] TH1A	–15°C/5°F...200°C/392°F	90°C/194°F
Thermal warning level for AI1 This parameter sets the temperature threshold to trigger the **[AI1 Th Warning]** TP1A warning. This parameter is visible when **[AI1 Th Monitoring]** TH1S is set to **[AI1]** AI1. Access path: **[Monitoring]** ➡ **[Thermal monitoring]**		

About [Current Limit] ILT

NOTE:

ILt on ATS48 is similar to **[Current Limit]** ILT on ATS480.

In general, using the same setting for ILT do not affect the starting behavior of your application.

Nevertheless, on ATS480, if a difference in the starting behavior has been detected, adjust the **[Current Limit]** ILT parameter by decreasing its value (1...10%).

Modbus Communication

What’s in This Part

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ATS480 Embedded Modbus Wiring

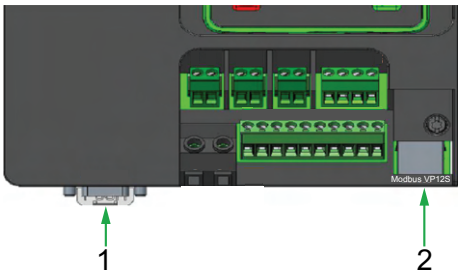
Compatible

It is possible to use the ATS48 communication architecture with the ATS480

- Use the same PLC, splitter boxes, and T-junction boxes.

The ATS48 communication wiring recommendations apply to the ATS480

- Use the same communication cable VW3A8306 on the embedded Modbus port.



1. Optional communication module slot
2. Modbus RTU VP12S

The following optional communication modules offer new communication protocols with the ATS480:

- Ethernet IP and Modbus TCP: VW3A3720 and VW3A3721
- Profibus DP: VW3A3607
- CANopen: VW3A3608, VW3A3618 and VW3A3628

Refer to the fieldbus manuals in Related Documents, page 13 for further instructions.

Modbus Configuration

Modbus Compatibility

Compatible

- **The ATS480 embedded Modbus is compatible with the ATS48 embedded Modbus.**
  - The parameter **[Control Mode]** CHCF must be set to **[SE8 Profile]** SE8 (factory setting).
  - Adapt to your installation the value set to **[Modbus Format]** TFO in the menu **[Communication]** COM ➡ **[Modbus Fieldbus]** MD1
  - The ATS48 Modbus addresses, ETA words, CMD words and frame format are functional with ATS480. No change required.
- **Use the configuration transfer tool, refer to** **Configuration Migration, page 45****.**
- **For a new configuration refer to the ATS480 User Manual on www.se.com.**

[Control Mode] CHCF

Parameter	Setting	Factory setting
[Control Mode] CHCF	–	**[SE8 Profile]** SE8

Access path: **[Complete settings]** ➡ **[Command channel]**

This parameter is relevant if the soft starter is used with a fieldbus.

- Set **[Control Mode]** CHCF to **[SE8 Profile]** SE8 to substitute the ATS48 fieldbus architecture. This setting enables the reuse of the same gateways, parameter mapping, command words and status words as ATS48.

  Available only in Modbus RTU.
- Set **[Control Mode]** CHCF to **[Standard Profile]** STD to use the latest evolutions of the embedded Modbus and the fieldbus modules.

  The **[Standard Profile]** STD is based on CIA402.

Plugging or unplugging a fieldbus module does not automatically change the value set in **[Control Mode]** CHCF . Set manually **[Control Mode]** CHCF to **[Standard Profile]** STD to use a fieldbus module.

To access this parameter you must set **[Access Level]** LAC in the menu **[My preferences]** MYP ➡ **[Parameter access]** PAC to **[Expert]** EPR .

HMIS status

Different HMIS status between ATS480 and ATS48 in Modbus command (**[Control Mode]** CHCF = **[SE8 Profile]** SE8) and *Stop command* = 1 and **[Mains Voltage]** ULN are present.

	Soft Starter ATS 48	Soft Starter ATS 480
CMD = 0	HMIS = NST	HMIS = NST
CMD = 6	HMIS = NST	HMIS = RDY
CMD = 15	HMIS = RUN	HMIS = RUN

Modbus Broadcast

	Soft Starter ATS 48	Soft Starter ATS 480
Client Configured at @248 (Broadcast)	off	off
Client Configured at @248 (Broadcast)	1	1
Client Configured at @1	1	1

	Doesn't answer to the client
	Answer to the client

Product HMI

What's in This Part

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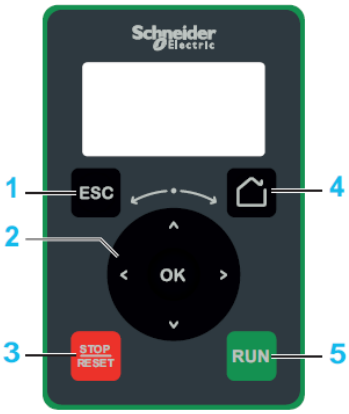
Mounting display terminal on the door of the enclosure 73
Soft Starter State 74
Front Product LEDs 75

Display Terminals

Plain Text Display Terminal VW3A1113

This Plain Text Display Terminal is a local control unit plugged on the soft starter. The Display Terminal can be removed to be mounted on the door of the wall-mounted or floor-standing enclosure, using a dedicated door-mounting kit, refer to Mounting display terminal on the door of the enclosure, page 73. The Display Terminal communicates with the soft starter using Modbus serial link. Both embedded Modbus connections (Modbus HMI & Modbus Fieldbus) can be used but only one Display Terminal must be connected at the same time.

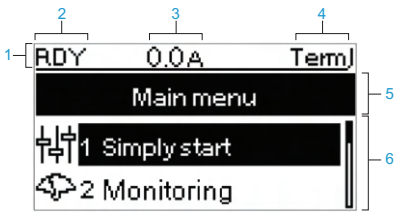


1. **ESC:** used to quit a menu/parameter, to clear the display of the triggered error or remove the currently displayed value in order to revert to the previous value retained in the memory
2. **Touch wheel / OK:** used to save the current value or access the selected menu/parameter. The touch wheel is used to scroll fast into the menus. Up/down arrows are used for precise selections, right/left arrows are used to select digits when setting a numerical value of a parameter.
3. **STOP / RESET:** stop command / apply a Fault Reset (a).
4. **Home:** used to access the home page.
5. **RUN:** executes the function (a).

(a) The function **RUN** of the **RUN** button and **RESET** of the **STOP / RESET** button are active only if the active command channel is the Display Terminal.

- **[Control Mode]** `CHCF` is set to **[Standard Profile]** `STD`
- **[Command Switching]** `CCS` is set to the channel commanding the display terminal

Example: Control via display terminal is active when **[Command Switching]** `CCS` is set to **[Cmd channel 1]** `CD1` and **[Cmd channel 1]** `CD1` is set to **[HMI]** `LCC`.



Key	
1	Display line.
2	Soft starter state, refer to Soft Starter State, page 74
3	Monitored parameter user defined. Can be configured in **[My preferences]**.
4	Active control channel • TERM: terminals • HMI: plain text display terminal • MDB: embedded Modbus serial • CAN: CANopen® • NET: Fieldbus module • PWS: DTM based commissioning software
5	Menu line: indicates the name of the current menu or submenu.
6	Menus, submenus, parameters, values, bar charts, and so on, are displayed in drop-down window format on a maximum of 2 lines. The line or value selected by the navigation button is displayed in reverse video.

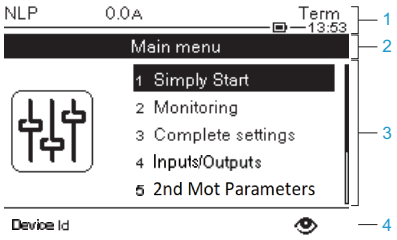
Graphic Display Terminal VW3A1111

The Graphic Display Terminal is available as an optional Display Terminal and can be plugged like the Plain Text Display Terminal, using the Modbus HMI serial link connection. This Display Terminal can also be mounted on the door of the wall-mounted or floor-standing enclosure, refer to Mounting display terminal on the door of the enclosure, page 73. Both embedded Modbus connections (Modbus HMI & Modbus Fieldbus) can be used but only one Graphic Display Terminal must be connected at the same time.

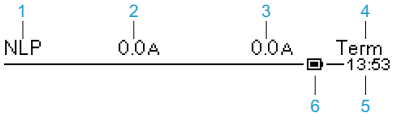
1. **STOP / RESET:** stop command / apply a Fault Reset (a).
2. **LOCAL / REMOTE:** used to switch between local and remote control of the soft starter. This button is disabled if **[HMI L/R cmd]** `BMP` is set to **[Disabled]** `DIS`, in the menu **[My preferences]** `MYP` ➡ **[Customization]** `CUS`.
3. **ESC:** used to quit a menu/parameter, to clear the display of the triggered error or remove the currently displayed value in order to revert to the previous value retained in the memory
4. **F1 to F4:** function keys used to access soft starter id, QR code, quick view, and submenus. Simultaneous press of F1 and F4 keys generates a screenshot file in the Graphic Display Terminal internal memory.
5. **Graphic display.**
6. **Home:** used to access the home page.
7. **Information:** used to have more information about menus, submenus, and parameters. The selected parameter or menu code is displayed on the first line of the information page.
8. **RUN:** executes the function (a).
9. **Touch wheel / OK:** used to save the current value or access the selected menu/parameter. The touch wheel is used to scroll fast into the menus. Up/down arrows are used for precise selections, right/left arrows are used to select digits when setting a numerical value of a parameter.
10. **RJ45 Modbus serial port:** used to connect the Graphic Display Terminal to the soft starter in remote control.
11. **MiniB USB port:** used to connect the Graphic Display Terminal to a computer.
12. **Battery:** The battery have no use for the soft starter and there is no alarm for display terminal low battery level.

(a) The function **RUN** of the **RUN** button and **RESET** of the **STOP / RESET** button are active only if the active command channel is the Display Terminal.

- **[Control Mode]** CHCF is set to **[Standard Profile]** STD
- **[Command Switching]** CCS is set to the channel commanding the display terminal



Key	
1	Display line.
2	Menu line: indicates the name of the current menu or submenu.
3	Menus, submenus, parameters, values, bar charts, and so on, are displayed in drop-down window format on a maximum of five lines. The line or value selected by the navigation button is displayed in reverse video.
4	Section displaying tabs (1 to 4 by menu), these tabs can be accessed using F1 to F4 keys



Key	
1	Soft starter state, refer to Soft Starter State, page 74.
2	Monitored parameter user defined, can be modified in **[My preferences]**.
3	Monitored parameter user defined, can be modified in **[My preferences]**.
4	Active control channel: • TERM: terminals • HMI: Graphic display terminal • MDB: integrated Modbus serial • CAN: CANopen® • NET: Fieldbus module • PWS: DTM based commissioning software
5	Present time.
6	Battery level. Level of battery embedded in the soft starter.

Graphic display terminal connected to a computer

The graphic display terminal is recognized as a USB storage device named SE_ VW3A1111 while plugged into a computer.

This allows to access the saved soft starter configurations (DRVCONF folder) and the graphic display terminal screenshots (PRTSCR folder).

Screenshots can be stored by a simultaneous press on F1 and F4 functions keys.

Mounting display terminal on the door of the enclosure

The ATS480 is delivered with the VW3A1113 Plain Text Display Terminal.

The VW3A1111 Graphic Display Terminal is available as an option to replace the Plain Text Display Terminal.

Door mounting kits are available as options to mount the display terminal on the door of the enclosure.

Refer to the following table to choose a display terminal and its door mounting kit.

Door Mounting Kit protection degree	Display terminal	Door mounting kit
IP43	VW3A1113 Plain Text Display Terminal **Delivered with the soft starter** 	VW3A1114 door mounting kit. **Available as option** Refer to the instruction sheet EAV91355.
IP65	VW3A1111 Graphic Display Terminal **Available as option** 	VW3A1112 door mounting kit. **Available as option** Refer to the instruction sheet EAV76406.
Select one of the following RJ45 cables to connect the remote mounting kit to the soft starter: • 1 meter: VW3A1104R10 • 3 meters: VW3A1104R30 • 5 meters: VW3A1104R50 • 10 meters: VW3A1104R100 **Not included with the remote kit**		

Soft Starter State

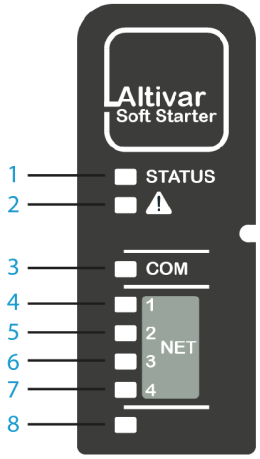
List of possible soft starter states, visible on the Display Terminal.

State	Condition
Displayed error label	Detected error. The soft starter is in operating state Fault.
Monitoring parameter selected by the user with the **[Display]** SUP menu. Factory setting: **[Motor Current]** LCR	Displayed value on the display terminal when the soft starter is running.
[Ready] RDY	No RUN command and mains supplied.
[No Mains Voltage] NLP	No RUN command and mains not supplied.
[Control Supply Loss] CLA	The warning **[Control Supply Loss]** CLA triggers when control supply is lost, the soft starter is not running and **[Control Supply Loss]** CLB is set to **[Warning]** 2
[Running] RUN	Soft starter running.
[Bypassed] BYP	Bypass active
[Accelerating] ACC	Soft starter in acceleration phase.
[Decelerating] DEC	Soft starter in deceleration phase.
[Wait for Restart] TBS	Starting time delay not elapsed.
[Operating State "Fault"] FLT	Detected error. The soft starter is in operating state Fault.
[Freewheel] NST	Soft starter forced to freewheel stop by serial link.
[Braking In Progress] BRL	Soft starter in braking phase.
[Cascade Waiting] STB	Waiting for a command (RUN or STOP) in cascade mode.
[Current Limitation] CLI	Soft starter in current limitation.
[Motor Preheating] HEA	Motor preheating, correspond to one of the following step of the preheating sequence: • Preheating order applied but **[Time Before Preheat]** TPR not elapsed, no preheating current injected yet • Preheating order applied and **[Time Before Preheat]** TPR elapsed, preheating current is injected
[Small Motor Test] SST	Small motor test in progress
[Firmware Update] FWUP	Firmware update mode
[Demo Mode] DEMO	Demonstration mode active

When current limitation is active, the displayed value flashes.

It is still possible to modify the parameters if the soft starter detects an error.

Front Product LEDs



Item	LED	Color & status	Description
1	**STATUS**	OFF	Indicates that the soft starter is not ready to start
		Green flashing	Indicates that the soft starter is not running, ready to start
		Green blinking	Indicates that the soft starter is in transitory status (acceleration, deceleration, and so on)
		Green on	Indicates that the soft starter is running
		Yellow on	Indicates that the soft starter localization is in progress
2	**Warning/Error**	Red flashing	Indicates that the soft starter has detected a warning
		Red on	Indicates that the soft starter has detected an error
3	**COM**	Yellow flashing	Indicates embedded Modbus serial activity
4	**NET 1**	Green/Yellow	For details, refer to the fieldbus manual
5	**NET 2**	Green/Red	
6	**NET 3**	Green/Red	
7	**NET 4**	Green/Yellow	
8	Reserved		

Glossary

A

AC:

Alternating Current

D

DC:

Direct Current

E

Error :

Discrepancy between a detected (computed, measured, or signaled) value or condition and the specified or theoretically correct value or condition.

F

Factory setting:

Machine status in factory settings when the product was shipped.

Fault Reset:

A function used to restore the soft starter to an operational state after a detected error is cleared by removing the cause of the error so that the error is no longer active.

Fault:

Fault is an operating state. If the monitoring functions detect an error, a transition to this operating state is triggered, depending on the error class. A "Fault reset" is required to exit this operating state after the cause of the detected error has been removed. Further information can be found in the pertinent standards such as IEC 61800-7, ODVA Common Industrial Protocol (CIP).

N

NC contact:

Normally Closed contact

NO contact:

Normally Open contact

O

OEM:

Original Equipment Manufacturer

OVCII:

Overvoltage Category II, according IEC 61800-5-1

P

PTC:

Positive Temperature Coefficient. PTC thermistor probes integrated in the motor or application to measure its temperature

W

Warning:

If the term is used outside the context of safety instructions, a warning alerts to a potential error that was detected by a monitoring function. A warning does not cause a transition of the operating state.

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As standards, specifications, and design change from time to time, please ask for confirmation of the information given in this publication.

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